

A norm-one conductor obstruction for weak elliptic curves

ISO-1 research note

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Abstract

For an odd extension of a finite field of size $q \equiv 1 \pmod{4}$, a Legendre parameter in the norm-one subgroup is a fourth power. Its rational 2-isogeny has a target with full rational 4-torsion. Thus one quadratic-twist sign has cardinality divisible by 16; in an ordinary isogeny class, this is equivalent to 4 dividing the Frobenius-order conductor. This proves the depth-one exclusion in the $q = p^2$, cubic-extension census at every odd prime, independently of the computed labels. We also prove that absolute Frobenius and inversion reduce that census to exactly $(p^4 + 3p^2 + 8)/12$ point-count calls for $p > 3$, with explicit exceptional-orbit weights.

1 The norm-one obstruction

Let $k = \mathbf{F}_q$, let $n \geq 3$ be odd, and put $K = \mathbf{F}_{q^n}$ and $Q = q^n$. Assume q is odd and $q \equiv 1 \pmod{4}$. Write $\sigma(z) = z^q$ and

$$T = \ker(N_{K/k} : K^\times \longrightarrow k^\times), \quad N = |T| = 1 + q + \cdots + q^{n-1}.$$

The generalized weak family is

$$E_\alpha : \quad y^2 = x(x - \alpha)(x - \sigma\alpha), \quad \alpha \in K \setminus k.$$

This is the cubic weak model used in the census when $q = p^2$ and $n = 3$. An affine root translation or a quadratic twist preserves the conclusion below whenever the resulting Legendre parameter belongs to T .

Theorem 1 (Norm-one conductor obstruction). *For every $\alpha \in K \setminus k$, put $\lambda = \sigma(\alpha)/\alpha$. Then $\lambda \in T \setminus \{1\}$, and the Legendre curve*

$$L_\lambda : \quad y^2 = x(x - 1)(x - \lambda)$$

has a rational 2-isogeny whose target has full rational 4-torsion. If $t = Q + 1 - \#E_\alpha(K)$, then

$$t \equiv \pm(Q + 1) \pmod{16}. \tag{1}$$

If E_α is ordinary, write

$$t^2 - 4Q = f_\pi^2 D_M,$$

where D_M is the fundamental discriminant of its imaginary quadratic endomorphism algebra and $f_\pi = [\mathcal{O}_M : \mathbf{Z}[\pi]]$. Then

$$4 \mid f_\pi, \quad v_2(f_\pi) \geq 2. \tag{2}$$

Proof. The norm of λ telescopes to 1. The integer N is odd, so raising to the fourth power is an automorphism of the cyclic group T . Choose $\mu \in T$ with $\lambda = \mu^4$ and put $s = \mu^2$. For $n = 3$ one can take

$$\mu = \lambda^{(q^2+q+2)/4}.$$

Here $\lambda \neq 0, 1$, since $\alpha \neq 0$ and $\sigma\alpha \neq \alpha$.

Set $a = -(1 + \lambda)$ and $b = \lambda$. On $L_\lambda : y^2 = x^3 + ax^2 + bx$, the map

$$(x, y) \mapsto \left(x + a + \frac{b}{x}, y \left(1 - \frac{b}{x^2} \right) \right) \quad (3)$$

for $x \neq 0$, with O and $(0, 0)$ mapped to O , is the rational 2-isogeny with kernel $\{O, (0, 0)\}$. Its target is

$$\begin{aligned} L'_\lambda : \quad y^2 &= x(x^2 - 2ax + (a^2 - 4b)) \\ &= x(x + (1 + s)^2)(x + (1 - s)^2). \end{aligned}$$

For $U = x^2 + ax + b$, substitution in the target equation is verified by

$$U^2 - 2axU + (a^2 - 4b)x^2 = (x^2 - b)^2.$$

The map extends across its poles on the nonsingular projective curves and sends O to O . Its x -coordinate has rational-function degree 2, so the resulting isogeny has degree 2 and the indicated kernel.

The three target roots are

$$e_0 = 0, \quad e_1 = -(1 + s)^2, \quad e_2 = -(1 - s)^2.$$

They are distinct because $\lambda \neq 0, 1$. Their differences are, up to sign, $(1 + s)^2$, $(1 - s)^2$, and $4s = (2\mu)^2$. As $Q \equiv 1 \pmod{4}$, -1 is a square in K , so every root difference is a square.

For completeness, the rational-halving calculation is explicit. On $y^2 = (x - e)(x - e + a_0^2)(x - e + b_0^2)$, the point

$$P = (e + a_0b_0, a_0b_0(a_0 + b_0))$$

lies on the curve and doubles to $(e, 0)$. The tangent slope is $a_0 + b_0$. Nonzero root differences ensure the denominators and $a_0 + b_0$ are nonzero. Apply this to each root of L'_λ . Halves of two independent 2-torsion points generate a copy of $(\mathbf{Z}/4\mathbf{Z})^2$. Thus $16 \mid \#L'_\lambda(K)$. This is the sufficient direction of the halving criterion in Auer–Top, Lemma 2.1 [1].

The substitution $x = \alpha X$, $y = \alpha Y$ puts E_α in the form $Y^2 = \alpha X(X - 1)(X - \lambda)$, a quadratic twist of L_λ . Isogenous curves have the same cardinality and quadratic twists have opposite traces. Therefore one of the two signs of t satisfies $Q + 1 \mp t \equiv 0 \pmod{16}$, proving (1). Lemma 2 gives (2). $\square$

Lemma 2 (Trace and conductor equivalence). *Let an ordinary elliptic curve over $\mathbf{F}_Q$, with $Q \equiv 1 \pmod{4}$, have Frobenius π , trace t , and conductor f_π . Then*

$$4 \mid f_\pi \iff t \equiv \pm(Q + 1) \pmod{16}.$$

Both twist signs have the same Frobenius-order conductor.

Proof. Suppose $\varepsilon t \equiv Q + 1 \pmod{16}$, with $\varepsilon \in \{1, -1\}$. The nonrational quadratic element

$$\eta = \frac{\varepsilon\pi - 1}{4}$$

has integral trace $(\varepsilon t - 2)/4$ and integral norm $(Q + 1 - \varepsilon t)/16$. Hence $\eta \in \mathcal{O}_M$, and

$$\text{disc } \mathbf{Z}[\eta] = \frac{t^2 - 4Q}{16} = \left(\frac{f_\pi}{4}\right)^2 D_M.$$

The conductor of the quadratic order $\mathbf{Z}[\eta]$ is an integer, so $4 \mid f_\pi$.

Conversely, write $\mathcal{O}_M = \mathbf{Z} + \mathbf{Z}\omega$ and $\pi = u + f_\pi v\omega$ with $u, v \in \mathbf{Z}$. If $4 \mid f_\pi$, the odd norm Q forces u to be odd. Choose ε with $\varepsilon u \equiv 1 \pmod{4}$. Then $(\varepsilon\pi - 1)/4 \in \mathcal{O}_M$; its integral norm gives $\varepsilon t \equiv Q + 1 \pmod{16}$. Finally $\mathbf{Z}[-\pi] = \mathbf{Z}[\pi]$. $\square$

Corollary 3 (The cubic census). *Let p be odd, $q = p^2$, and $K = \mathbf{F}_{p^6}$. Every ordinary trace represented by the cubic weak family satisfies*

$$t \equiv \pm(p^6 + 1) \pmod{16}, \quad v_2(f_\pi) \geq 2.$$

These conclusions hold for every p , including primes beyond the completed censuses.

2 The endomorphism orders on the edge

Corollary 4. *Let L_λ and L'_λ in Theorem 1 be ordinary. If f_E and $f_{E'}$ are the conductors of their endomorphism orders, then*

$$f_{E'} \mid f_\pi/4, \quad v_2(f_E) \leq v_2(f_\pi) - 1.$$

The latter inequality also holds for every ordinary curve with full rational 2-torsion; the full-4-torsion neighbor is the additional structure supplied by the norm-one construction.

Proof. On the target, $\pi - 1$ kills $E'[4]$. Since $[4]$ is separable, $\pi - 1$ factors through $[4]$, giving $(\pi - 1)/4 \in \text{End}(E')$. Its generated order has conductor $f_\pi/4$, so $f_{E'}$ divides that integer.

Identify the two quadratic endomorphism algebras through the isogeny $\phi : E \rightarrow E'$ and its dual $\hat{\phi}$, with $\hat{\phi}\phi = [2]$. Conjugation by ϕ shows $2\text{End}(E) \subseteq \text{End}(E')$ and $2\text{End}(E') \subseteq \text{End}(E)$. Writing each quadratic order as $\mathbf{Z} + f\mathcal{O}_M$ gives $f_E \mid 2f_{E'}$ and $f_{E'} \mid 2f_E$. Their 2-adic conductor depths therefore differ by at most one. For any curve with full rational 2-torsion, the same factorization with $[2]$ gives $(\pi - 1)/2 \in \text{End}(E)$ and $f_E \mid f_\pi/2$. $\square$

3 An exact absolute-Frobenius orbit count

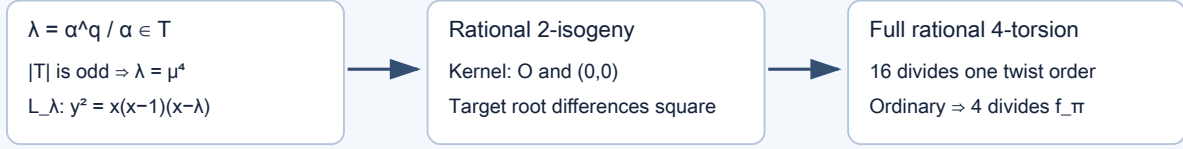
Lemma 5 (Parameters and twist signs). *For $K = \mathbf{F}_{p^6}$ and $k = \mathbf{F}_{p^2}$, the map $\alpha \mapsto \alpha^{p^2-1}$ induces a bijection*

$$K^\times/k^\times \longrightarrow T.$$

Removing $\lambda = 1$ removes precisely the base-field line. For each remaining parameter, the two square classes of scaling by $k^\times$ give opposite trace signs.

Norm-one parameter $\rightarrow$ conductor obstruction

$q \equiv 1 \pmod{4}$, odd $n \geq 3$, $K = F_-(q^n)$, $Q = q^n$



Theorem applies at every size; independent of the census point counter.

Cubic extension $q = p^2$: exact absolute-Frobenius quotient

$\lambda \mapsto \lambda^p$ and $\lambda \mapsto \lambda^{-1}$ preserve the trace pair. Orbit sizes: 2, 6, 12.

One point count per orbit: $C_p = (p^4 + 3p^2 + 8)/12$ for $p > 3$.

Figure 1: The proved implication and the orbit reduction used in the larger-prime continuation. The census labels test the remaining class-existence question.

Proof. The kernel is $k^\times$, and the image has cardinality $(p^6 - 1)/(p^2 - 1) = |T|$, so it is all of T . A nonsquare in k remains a nonsquare in the cubic extension, since the exponent $1 + p^2 + p^4$ is odd. Scaling α by it produces the quadratic twist. Thus a census of normalized parameters and both scaling classes has $2(p^4 + p^2)$ representatives. $\square$

Theorem 6 (Absolute-Frobenius quotient). *For a prime $p > 3$, the action on $T \setminus \{1\}$ generated by $\lambda \mapsto \lambda^p$ and $\lambda \mapsto \lambda^{-1}$ preserves the trace pair. Its orbit counts are*

orbit size	number of parameters	number of orbits
2	2	1
6	$2(p^2 - 1)$	$(p^2 - 1)/3$
12	$p^4 - p^2$	$(p^4 - p^2)/12$.

Consequently the exact number of point-count calls needed for one representative per orbit is

$$C_p = \frac{p^4 + 3p^2 + 8}{12}. \quad (4)$$

Weighting each result by its orbit size and adding its twist sign recovers the normalized-representative census.

Proof. The absolute Frobenius bijects rational points and preserves T . Inversion preserves the Legendre cardinality because λ is a square: with $r^2 = \lambda$, the map $(x, y) \mapsto (x/\lambda, y/(\lambda r))$ gives $L_\lambda \simeq L_{1/\lambda}$ over K . The two operations commute and give the twelve exponents $\pm p^j$, $0 \leq j < 6$.

Put $A = p^2 + p + 1$ and $B = p^2 - p + 1$. Then $|T| = AB$, $\gcd(A, B) = 1$, $p^3 \equiv 1 \pmod{A}$, and $p^3 \equiv -1 \pmod{B}$. The subgroups T_A, T_B have trivial intersection. Their nonidentity union has $A + B - 2 = 2p^2$ elements.

The fixed-point equation for a transformation is $\lambda^{\pm p^j - 1} = 1$. The identities

$$\gcd(AB, p^3 - 1) = A, \quad \gcd(AB, p^3 + 1) = B, \quad \gcd(AB, p^2 - 1) = 3, \quad \gcd(AB, p^2 + 1) = 1$$

and $\gcd(AB, p \pm 1) \mid 3$ classify all stabilizers; $p^6 \equiv 1 \pmod{AB}$ reduces $j = 4, 5$ to $j = 2, 1$. The two primitive cube roots form one orbit of size 2. The other $2p^2 - 2$ elements in $T_A \cup T_B$ have orbit

size 6. All remaining $p^4 - p^2$ elements have trivial stabilizer and orbit size 12. Summing the orbit counts gives (4). Lemma 5 supplies both scaling classes and their trace signs. $\square$

p	Normalized representatives	Six-element quotient	C_p
13	57,460	4,789	2,423
37	3,751,060	312,589	156,523
47	9,763,780	813,649	407,193
53	15,786,580	1,315,549	658,243
59	24,241,684	2,020,141	1,010,651
61	27,699,124	2,308,261	1,154,751
199	3,136,557,604	261,379,801	130,696,501

These are exact counts of parameters and point-count calls. Execution times depend on the field kernel, the point counter, and host conditions.

4 Evidence and the next class-existence test

The original rational-isogeny factorization and cleared equation have native identity records

$$\text{IDC1h8ae7209125dfe0a2} \quad \text{IDC1he02ab9a5cf019a56}.$$

Separate exact polynomial expansion is performed in PARI/GP. The additional halving and orbit arithmetic have records

$$\text{IDC1hf8aba4800a6ceb5c} \quad \text{IDC1h1bc37ec40bcb8619}.$$

The generalized PARI controls passed 1,100 norm-one cases and all 476 ordinary Hasse traces tested for the conductor equivalence. The companion validation receipt records their frozen sources and the larger-prime census status. The proofs above do not depend on randomized point-count assignments.

The exact $p = 7$ census counts every field element for each of the 213 absolute orbits and agrees, including all weights, with an independent complete PARI census. It finds 126 weak ordinary rows among 294, with 20 zero rows among the 146 ordinary rows of depth at least two. In particular the ordinary class $t = 610$ has

$$Q = 117,649, \quad t^2 - 4Q = -98,496 = 72^2(-19), \quad v_2(f_\pi) = 3,$$

and has no norm-one weak representative. The independent existence check realizes this trace with three rational 2-torsion roots; the quadratic twist gives $t = -610$. Together with the exhaustive norm-one enumeration, this supplies a finite, certified counterexample to sufficiency of the depth condition for the cubic weak family.

The completed diagnostic censuses at $p = 37, 41, 43, 47$, and 53 contain, respectively, 290, 396, 376, 424, and 494 ordinary zero-witness rows at conductor depth at least two. Those rows remain labels from the retained point counter.

The $p = 53$ run has 72,540 weak ordinary rows among 146,068; all 5,000 independent PARI controls are in its positive set.

The next existence test is to certify those labels and test invariants of the order containing $(\pi - 1)/4$, while extending the frozen protocol toward $p = 199$. A class criterion needs a sufficient argument for the norm-one parameter condition as well as the conductor obstruction.

References

- [1] R. Auer and J. Top, *Legendre elliptic curves over finite fields*, 2001, Lemma 2.1 and Proposition 2.1. <https://arxiv.org/abs/math/0106273>.
- [2] A. Joux and V. Vitse, *Cover and Decomposition Index Calculus on Elliptic Curves Made Practical*, EUROCRYPT 2012. <https://eprint.iacr.org/2011/020>.
- [3] The PARI Group, *PARI/GP function reference: elliptic curves*, entries `ellcard` and `ellgroup`. https://pari.math.u-bordeaux.fr/dohtml/html/Elliptic_curves.html.